

Exercise 3 a) A Brief Introduction into C-Programming (for Java-Programmers)

Learning objectives:

- Distinguish the Differences in C and Java (development and execution, pointer, memory allocation and management)
- Developing C-Programms by a typical IDE

Legend:

 Reading to be completed before the exercise

 Questions/tasks to be completed before the exercise

 Tasks to be worked on during the exercise

3.1 Tool Preparation

If you decide to use Eclipse for C-Programming, there are some useful hints especially for MacOS and Windows in <https://www.cs.rochester.edu/u/ferguson/csc/c/eclipse-c.html>.

For Linux, you could obtain `eclipse` and the C-Compiler `gcc` plus `tools` like `make` from your distribution provider directly. For ubuntu take the following information:

- `sudo apt install gcc`
- `sudo apt install make`
- <https://linuxconfig.org/eclipse-ide-for-c-c-developers-installation-on-ubuntu-22-04>
- <https://wiki.ubuntuusers.de/Eclipse/>

The Raspberry Pi Computers already have all tools installed. No further work is needed.

If you decide for another IDE or editor, it is possible. Though, it is your own decision and no additional support can be given.

3.2 Exercise Preparation

The preparations and questions should be completed before the exercise session.

 Study the following materials for preparation. Concentrate on Chapters 2, 3, 4, 5, 6, 8, and 10. The Chapters 7 and 11 are optional.

- <https://www.cs.rochester.edu/u/ferguson/csc/c/c-for-java-programmers.pdf>



Introductory Questions:

What are the difference between C and Java? Please tick all differences:

- ☐ Object Oriented Concepts
- ☐ Pointer
- ☐ Memory Management
- ☐ Execution Environment
- ☐ Compilation Process
- ☐ Numerical Datatypes
- ☐ String handling

What is a pointer in C?

How do you get an address of a variable in C? Name the operator and its symbol.

How are arguments of functions passed in C (like in Java)?

How do you call the fact, if an argument of a function could be changed within the function in C and the change is available outside of the function after its call? What is used for this in C-Language?

Look at the C program in Listing 2 in the following subsection 3.4. What do you expect to be printed while executing the program?

Before calling swap: number1 = , number2 =
After calling swap: number1 = , number2 =
Return value from swap:

3.3 Compiling and Executing a C-Programm

❖ Create a new project **helloWorld** in eclipse. Add a new C-source **helloWorld.c** file to this project. Copy the following statements from Listing 1 into the file.

Listing 1: helloWorld.c

```
/*  
 * File: helloWorld.c  
 */  
#include <stdio.h>  
  
int main()  
{  
    printf("Hello World!\n");  
}
```

Compile, link and run the program.

3.4 Passing arguments by value in a C-Programm

❖ Given is a C-Program in Listing 2 which contains a `main` function and a `swap` function. The `swap` function is called using two arguments `a` and `b`. The function returns a single value, which is assigned to the variable `returnval`.

Listing 2: `swapfunction.c`

```
/*
 * swapfunction.c
 *
 *      Author: abke
 */

#include <stdio.h>

int swap(int a, int b)
{
    int    tmp = 0;

    tmp = a;
    a = b;
    b = tmp;

    return(a);
}

int main()
{
    int    number1    = 10;
    int    number2    = 5;
    int    returnval  = 0;

    printf("Before calling swap: number1 = %d, number2 = %d\n",
           number1, number2);

    returnval = swap(number1, number2);

    printf("After  calling swap: number1 = %d, number2 = %d\n",
           number1, number2);

    printf("Return value from swap: %d\n", returnval);

    return(0);
}
```

Take the program and copy it to a new project **reallyswapfunction**. Modify the copied program in that way, that the values of the variables a and b are swapped after the call of the function swap.

Compile, link and run the program. What is the output of the program?

3.5 Printing values

❖ There is a fragment of a C-Program given in Listing 3 which contains some variables defined. Some values are assigned.

Listing 3: printingValues.c

```
/*
 * printingValues.c
 *
 *      Author: abke
 */

int main()
{
    int    number    = 42;
    long   empty;
    float  value     = 17.5;
    char   string[6] = "Hello";

    /* Add the print function calls for number, empty and value here */

    /* Print out the string variable in complete */

    /* Print out character by character of the string variable */

    return(0);
}
```

Extend the program by adding respective function calls to print all values of the variables. Use the respective function call line by line. All outputs should be in separate lines. Additionally, print out the variable **string** completely and character by character.